

Second-Chance Page Replacement Algorithm

1 Core Idea

Definition

The **second-chance page replacement algorithm** is a modified FIFO algorithm that checks the referenced bit R before evicting the oldest page.

FIFO removes the oldest page even if it is still heavily used. Second chance improves this by giving a recently referenced page one more chance to stay in memory.

Second-Chance Rule

$R = 0$: evict the page
 $R = 1$: clear R and move the page to the tail

2 How It Modifies FIFO

The operating system keeps pages in FIFO order:

head = oldest page, tail = newest page

On a page fault, it examines the page at the head of the list.

State	Action
$R = 0$	The page is old and not recently used, so evict it.
$R = 1$	Clear R , move the page to the tail, update its load time, and continue searching.

Main Intuition

An old page should not be evicted immediately if it was referenced recently.

3 Algorithm Steps

1. A page fault occurs.
2. Look at the oldest page, at the head of the FIFO list.
3. If its R bit is 0, choose it as the victim.
4. If its R bit is 1, clear R .
5. Move that page to the tail of the list, as if it just arrived.
6. Repeat the search with the next oldest page.

4 Example: Page A Gets a Second Chance

Setup

Pages A through H are in memory in FIFO order. The numbers are their load times.

Head $A_0, B_3, C_7, D_8, E_{12}, F_{14}, G_{15}, H_{18}$ Tail

A page fault occurs at time 20.

The oldest page is A , loaded at time 0.

Case	Result
$A.R = 0$	A has not been referenced recently, so evict A .
$A.R = 1$	Clear $A.R$, move A to the tail, and set its new load time to 20.

If $A.R = 1$, the list becomes:

Head $B_3, C_7, D_8, E_{12}, F_{14}, G_{15}, H_{18}, A_{20}$ Tail

Meaning

Page A was old, but it was recently referenced. Second chance delays its eviction and tests page B next.

5 What the Algorithm Is Looking For

Second chance searches for an old page that has not been referenced in the most recent clock interval.

Page type	How second chance treats it
Old and $R = 0$	Good victim. Evict it.
Old but $R = 1$	Recently used. Clear R and move it to the tail.

6 If All Pages Have $R = 1$

Degenerates to FIFO

If every page has $R = 1$, second chance eventually becomes pure FIFO.

Here is why it still terminates:

1. The algorithm starts at the oldest page.
2. Each page has $R = 1$, so each page is moved to the tail and has R cleared.
3. Eventually the algorithm returns to the original oldest page.
4. That page now has $R = 0$, so it is evicted.

Guarantee

Second chance always finds a victim because it clears R bits while scanning.

7 Strengths and Weaknesses

Point	Explanation
Strength	Simple improvement over FIFO.
Strength	Avoids immediately evicting recently used old pages.
Weakness	May scan many pages before finding a victim.
Weakness	If all pages were referenced, it behaves like FIFO.

8 Final Summary

One-Box Summary

Second chance is FIFO plus an R -bit check.
If $R = 0$, evict the oldest page.
If $R = 1$, clear R and move the page to the tail.
It protects recently used pages, but can degrade to FIFO.